Differential Equations

By Naren Ayinala

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1 Basic Definitions

A **differential equation** relates a dependent variable, an independent variable, and derivatives of the dependent variable.

$$\frac{dy}{dx} + y = x, \qquad \frac{d^2y}{dx^2} + 3\frac{dy}{dx} = 0.$$

Order: the order of the highest derivative present. **Degree:** the power of the highest-order derivative after the equation has been expressed as a polynomial in derivatives (when possible).

$$\left(\frac{d^2y}{dx^2}\right)^3 + \left(\frac{dy}{dx}\right)^2 = 0 \quad \Longrightarrow \quad \text{order } 2,\ \text{degree } 3.$$

2 General and Particular Solutions

A **general solution** contains arbitrary constants. A **particular solution** results when initial or boundary conditions determine those constants.

$$\frac{dy}{dx} = 2x \quad \Longrightarrow \quad y = x^2 + C.$$

If $y(1) = 4$, then $4 = 1 + C$, hence $\boxed{y = x^2 + 3}$.

3 Formation of a Differential Equation

To form a differential equation from a family of curves, differentiate as many times as needed and eliminate arbitrary constants. **Example.** Given $y = Ce^{2x}$, differentiate:

$$\frac{dy}{dx} = 2Ce^{2x} = 2y \quad \Longrightarrow \quad \boxed{\frac{dy}{dx} - 2y = 0}.$$

For $y = mx + c$, differentiating twice gives $\boxed{d^2y/dx^2 = 0}$.

4 Variable-Separable Equations

If $\dfrac{dy}{dx} = f(x)g(y)$, rearrange and integrate:

$$\boxed{\int \frac{1}{g(y)}\,dy = \int f(x)\,dx + C.}$$

The rearrangement assumes $g(y) \neq 0$; check constant solutions satisfying $g(y) = 0$ separately. **Example.** $dy/dx = xy$:

$$\frac{dy}{y} = x\,dx \implies \ln|y| = \frac{x^2}{2} + C \implies \boxed{y = Ce^{x^2/2}}.$$

5 Homogeneous Differential Equations

A first-order equation $dy/dx = F(y/x)$ is homogeneous. Substitute $y = vx$, so

$$\boxed{\frac{dy}{dx} = v + x\frac{dv}{dx}}.$$

The resulting equation is separable. Alternatively, for $dx/dy = G(x/y)$, use $x = vy$. **Example.** Solve $dy/dx = 1 + y/x$ ($x \neq 0$):

$$v + x\frac{dv}{dx} = 1 + v \implies x\frac{dv}{dx} = 1 \implies v = \ln|x| + C.$$

Thus $\boxed{y = x(\ln|x| + C)}$.

6 Linear Differential Equations

The standard first-order linear equation is

$$\boxed{\frac{dy}{dx} + P(x)y = Q(x)}.$$

Its **integrating factor** is

$$\boxed{\text{IF} = e^{\int P(x)\,dx}}.$$

Multiply by IF and integrate:

$$\boxed{y\,\text{IF} = \int Q(x)\,\text{IF}\,dx + C}.$$

Example. $dy/dx + 2y = e^x$ has IF e^{2x}, hence

$$ye^{2x} = \int e^{3x}\,dx = \frac{e^{3x}}{3} + C \implies \boxed{y = \frac{e^x}{3} + Ce^{-2x}}.$$

For $dx/dy + P(y)x = Q(y)$, use IF $= e^{\int P(y)\,dy}$ and solve analogously.

7 Quick Identification Guide

Type	Method
$dy/dx = f(x)g(y)$	Separate variables
$dy/dx = F(y/x)$	Put $y = vx$
$dy/dx + Py = Q$	Integrating factor
$dx/dy + Px = Q$	Integrating factor in y

Always differentiate the final answer to verify it satisfies the original equation.